Registro de Calibración del Software (según IEC 61083-2)

1. Información de la Validación

Software Probado: Analizador de Impulsos atmosféricos tipo rayo (1.2/50 µs). (v1.0.0) - 2026-03-24

Algoritmos soportados: Full Lightning Impulse (LI), Chopped Lightning Impulse (LIC)

Parámetros validados: Valor Pico (Ut), Tiempo de Frente (T1), Tiempo de Cola (T2), Sobrepasamiento (OS)

Generador de Datos de prueba (TDG): IEC 61083-2 TDG Version 2.1.0

Resolución TDG: 8 bits | Tasa de Muestreo: 100 MSa/s

2. Tabla de Resultados

2.1 Full Lightning Impulses - LI

Onda	Ut [kV]			T1 [µs]			T2/Tc [µs]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LI-A1	1049.60	1050.23	0.060%	0.840	0.840	-0.048%	60.16	60.22	0.097%	0.00	0.18	0.184%
LI-A2	1037.60	1038.00	0.039%	1.693	1.698	0.275%	47.48	47.45	-0.062%	5.10	4.77	-0.332%
LI-A3	1000.20	1001.97	0.177%	1.117	1.121	0.315%	48.15	48.25	0.213%	4.60	4.28	-0.324%
LI-A4	856.01	856.57	0.065%	0.841	0.855	1.621%	47.80	47.77	-0.055%	7.90	7.39	-0.512%
LI-A5	71.97	72.01	0.053%	1.711	1.720	0.510%	47.71	47.68	-0.061%	7.70	7.00	-0.698%
LI-A6	100.17	100.25	0.078%	1.762	1.764	0.107%	41.58	41.49	-0.206%	17.70	15.90	-1.796%
LI-A7	104.35	104.42	0.070%	2.122	2.125	0.133%	38.36	38.34	-0.045%	20.10	17.70	-2.396%
LI-A8	96.01	96.09	0.081%	1.503	1.508	0.338%	44.92	44.83	-0.200%	14.80	13.46	-1.339%
LI-A9	55.93	55.95	0.038%	1.215	1.222	0.537%	55.74	55.74	-0.008%	4.00	3.66	-0.340%
LI-A10	81.93	82.02	0.108%	0.924	0.931	0.742%	42.66	42.59	-0.171%	12.00	11.16	-0.836%
LI-A11	86.60	86.64	0.048%	0.578	0.592	2.367%	56.37	56.37	-0.002%	4.10	3.85	-0.255%
LI-A12	85.58	85.62	0.038%	0.587	0.592	0.817%	57.36	57.35	-0.011%	2.30	2.09	-0.212%
LI-M1	952.09	953.01	0.097%	1.123	1.131	0.707%	85.60	85.69	0.109%	2.10	1.89	-0.211%
LI-M2	-1041.70	-1041.54	-0.015%	3.356	3.358	0.051%	61.25	61.34	0.146%	9.20	8.75	-0.448%
LI-M3	-1026.50	-1026.44	-0.006%	2.150	2.153	0.162%	41.75	41.82	0.165%	9.20	8.69	-0.506%
LI-M4	-267.14	-267.15	0.003%	0.987	0.989	0.237%	56.22	56.17	-0.083%	4.80	4.51	-0.294%
LI-M5	-55.00	-55.00	0.003%	2.746	2.750	0.134%	42.11	42.09	-0.054%	18.70	16.38	-2.318%
LI-M6	-166.87	-166.88	0.005%	1.356	1.355	-0.073%	54.74	54.73	-0.011%	3.80	4.10	0.298%
LI-M7	-1272.30	-1272.88	0.045%	1.482	1.492	0.647%	50.03	50.02	-0.021%	11.20	10.81	-0.388%
LI-M8	-99.73	-99.81	0.077%	1.515	1.518	0.208%	49.36	49.27	-0.177%	-0.50	-0.30	0.199%
LI-M9	-100.04	-100.07	0.033%	0.828	0.829	0.094%	46.65	46.74	0.183%	1.40	1.20	-0.199%
LI-M10	100.26	100.26	0.002%	1.666	1.663	-0.166%	60.85	60.86	0.016%	0.00	-0.01	-0.009%
LI-M11	299.32	299.30	-0.007%	1.661	1.658	-0.210%	60.95	60.84	-0.178%	-0.50	-0.10	0.398%
LI-M12	-4.32	-4.32	-0.085%	1.292	1.282	-0.785%	52.27	52.40	0.248%	-1.80	-1.55	0.247%
LI-M13	39.46	39.53	0.189%	1.537	1.544	0.443%	46.94	46.97	0.073%	1.80	1.37	-0.433%
LI-M14	48.55	48.49	-0.124%	0.933	0.940	0.738%	37.48	37.35	-0.343%	4.30	3.68	-0.622%
LI-M15	497.97	497.63	-0.068%	1.017	1.017	0.020%	59.19	59.18	-0.013%	-0.10	-0.19	-0.090%
LI-M16	369.21	369.46	0.069%	0.920	0.921	0.107%	47.53	47.55	0.044%	0.80	0.76	-0.035%
LI-M17	-99.35	-99.23	-0.120%	1.775	1.772	-0.167%	53.31	53.18	-0.239%	1.30	1.34	0.038%

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2.2 Chopped Lightning Impulses (LIC)

Onda	Ut [kV]			T1 [µs]			T2/Tc [µs]			OS [%]		
	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv	Ref	Calc	Desv
LIC-M4	0.15	0.15	0.080%	1.305	1.310	0.358%	6.00	6.01	0.134%	-0.20	0.69	0.890%
LIC-M5	-389.90	-389.54	-0.092%	0.857	0.870	1.499%	9.24	9.27	0.305%	6.80	7.04	0.243%

3. Estimación de Incertidumbre (según Anexo B.3.3)

Basado en las máximas desviaciones encontradas:

3.1 Full Lightning Impulses - LI

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.1092	0.0150	**0.1102**
Tiempo de Frente (T1)	1.3666	0.2500	**1.3893**
Tiempo de Cola (T2/Tc)	0.1981	0.0300	**0.2003**
Sobrepasamiento (OS)	1.3835	0.0400	**1.3841**

3.2 Chopped Lightning Impulses (LIC)

Parámetro	u_B71 (Software) [%]	u_B72 (Referencia) [%]	u_B7 (Combinada) [%]
Valor Pico (Ut)	0.0531	0.0250	**0.0587**
Tiempo de Frente (T1)	0.8654	0.4500	**0.9754**
Tiempo de Cola (T2/Tc)	0.1763	0.1500	**0.2315**
Sobrepasamiento (OS)	0.5137	0.0250	**0.5144**